

EBE06 Template for Corrections to a Master's Dissertation

Note: This template is used by a Master's candidate to list the corrections made to their dissertation in response to the examiners' comments/ criticisms

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Dissertation Title	Design and Implementation of Tracking Filters for a Passive Radar System

	EXAMINER'S RECOMMENDATION/COMMENT	STUDENT'S RESPONSE	Page No affected
1	In section 1.2 ADS-B is described as the most common technology for air traffic control. Secondary surveillance radar is probably more common and should be mentioned here.	Section 1.2 has been revised and now includes ADS-B and Secondary Surveillance Radar as some of the most common technologies for air traffic control and are explained in detail in Section 1.2 Paragraph 1 and 2.	4 and 5
2	Section 2.3.1 describes basic elements and problems of conventional multi-target trackers. Many of these are also found in single target trackers. It may be beneficial to show which are unique to multitarget trackers	The figure has been updated to highlight elements unique to multi-target tracking, and the paragraph following the figure has also been updated.	28

3	At the bottom of page 74 additional peaks in the range-Doppler map are ascribed to sidelobe returns. Are these antenna sidelobes? It is not clear how this is possible.	The additional peaks observed in the range-Doppler map arise from the ambiguity function of the FM waveform used and persist regardless of the window function applied during range-Doppler processing and have been clarified in the first paragraph on page 76.	76
4	Data in Figures 4.5 and 4.6 appear to have been normalized relative to the peak levels in each figure. It would be interesting to know what the values were before this normalization, since there is no evidence of a target in the first figure	The figures 4.5 and 4.6 have been updated with the normalization removed. The figures are now 4.6 and 4.7.	77, 78
5	Where do the artefacts in figure 4.6 come from? They do not coincide with peaks in figure 4.5.	Since ECA has been applied, some returns that were previously masked by the DPI are now visible; these returns had lower magnitudes and did not appear in Figure 4.6 (previously 4.5). Additionally, certain artefacts present in Figure 4.6 were caused by the DPI itself and have been removed following DPI cancellation. The artefacts in Figure 4.7 (previously 4.6) have a different origin and do not coincide with the peaks in Figure 4.6.	
6	Does figure 4.12 follow from the previous results? If so, where do the extra detections come from?	Yes, Figure 4.12 follows from the previous results. The additional detections arise because they were captured at a different time during the simulations. The CA-CFAR and centroiding plots have now been updated to show detections at the same time for consistency. These are now presented as Figures 4.11 and 4.13.	82,84

7	It appears as though the measured data for figure 4.14 a) and b) are not the same. That makes it difficult to compare the results.	The differences in the measured data were due to an error. The figures have been updated as Figures 4.18 a and b (previously 4.14 a and b) and corrected to use the same measured data.	92
8	In figure 5.4 and 5.5, the y-axis scale is very small and difficult to read. It may be better to use $\mu\mu ss$ as a unit.	Figures 5.4 and 5.5 have been updated to use μs as the unit on the y-axis, and the axis fonts have been increased for better readability.	105
9	In Figure 5.13 it appears as though the particle filter has the smallest absolute tracking error in large areas of the maneuver, while the log likelihood calculations show that it has the worst performance. Is this correct?	Yes, this is correct. The log-likelihood evaluates how likely the predicted observations are given the actual ground truth, while accounting for uncertainties in the system (the filter's covariance or particle distribution). The particle filter may have some particles close to the true state with large weights, resulting in a low absolute tracking error, but the overall particle distribution may not properly represent the true posterior uncertainty, leading to poorer log-likelihood performance.	
10	On page 114 above the table, it is mentioned that the scenario includes two targets, while below the table it is stated that there are three targets.	This is an error. The scenario includes three targets, and the text above the table has been corrected accordingly in the last paragraph on page 120	120
11	Introduction: The introduction effectively establishes the background, problem statement, aims and scope, but, as a non-specialist in radar, there are some initial explanations of important concepts that should be improved.	The contrast between Active Radar and Passive Radar has now been highlighted in the first description on page 1 paragraph 2.	1

	• Passive vs. traditional radar: the most critical aspect – that passive radar does not involve a dedicated transmitter and instead uses transmission signals from external sources – does not clearly come across in this first description.		
12	• Surveillance and reference signals: state clearly that the reference signal is a reading of the external transmission signal, while the surveillance signal is the reflection from the target.	The definitions of the surveillance and reference signals have now been expanded to clarify that the reference signal is a direct line-of-sight recording of the illuminator signal, while the surveillance signal contains the target reflections along with the direct-path signal and other interference in paragraph 1 of section 1.1.	2
13	• Direct path interference: rather than just stating that it is the transmission signal, it is important to mention that the interference is caused by this signal directly travelling to the receiver without bouncing off targets in between.	The definition for the DPI has been improved in the last paragraph on page 3.	3
14	• Constant false alarm rate (CFAR) map: give a brief one or two sentence description elaborating on what this is and how it is obtained. It is not obvious whether it is an algorithm, a probability distribution, a collection of measurements or some other form of information, or what its inputs and outputs are. “More signal processing is required to process detections from the CFAR to tracks where targets can be visually followed, such as centroiding...” – the terminology that follows is not explained until much later in the dissertation, so it would be better to give more high-level descriptions e.g. instead of centroiding, “clustering of points into individual targets”, etc.	CFAR has now been clarified and explained in detail, and further signal processing has been described in paragraph 2 on page 4.	4
15	• Bistatic range, doppler domain: these concepts are very important to defining the scope of the project, but they are not	The Bistatic range and Doppler domain have now been	4

	clearly and explicitly defined anywhere in the report. They should be explained somewhere early in the report.	explicitly defined in the overview section, in paragraph 1 on page 4.	
16	Figure 1.1: The fonts on this plot are almost illegibly tiny. This is true of many figures in the dissertation, so the student should go through them all and ensure that no fonts in the legends, labels or elsewhere have an effective size of smaller than 10 pt when resized and placed into the final document.	The fonts for all the plots have been checked and resized where necessary in Figure 1.1, 4.12, 4.18, 4.20, 4.22, 4.27, 5.1, 5.2, 5.3, 5.4, 5.5, 5.8, 5.9, 5.10, 5.13, 5.14, 5.18, 5.19, 5.20 and 5.21	3, 82, 92, 93, 95, 98, 101, 102, 103, 105, 109, 110, 112, 113, 115, 116, 117 and 120
17	The ARD is a very specific visualization to this field, so it would be helpful to have more information on how to interpret this plot. What do the measurements on the axes mean in terms of the real-world movement or position of the target relative to the antennae? What are the important observable differences between the left and right plots and what do they mean?	The explanation of bistatic range and Doppler has been expanded to clarify real-world target movement, and the differences between the left and right plots have also been noted in the last paragraph on page 3 and paragraph 1 on page 4.	3,4
18	“FM radio signals provide low-range resolution...” this typo appears more than once, and it makes the meaning confusing. Is the range resolution low? Or is the resolution at low frequency ranges?	This is a typo. It was intended to indicate that the range resolution is low, and it has been corrected to “poor range resolution” in paragraph 2 on page iii and the last paragraph on page 5.	lii,5
19	Literature Review: The literature review is decently comprehensive and the approaches discussed are generally explained concisely and effectively brought back to the context of the project. Like the introduction, however, there are some initial explanations of ideas here that could be explained in slightly more detail for non-specialist readers. • What is it that made the Kalman filter so revolutionary? Give a brief description of the technical feature(s) of this filter that differentiated it from approaches at the time. The Kalman filter	The literature section on the Kalman filter has been revised to include more details on the filter mechanism and what made it revolutionary in section 2.2.1 paragraph 1 and 2.	17,18

	subsection could also be improved by a more detailed explanation of the mechanism of this filter in the first paragraph, so its place in the space of methods and its limitations, can be better understood.		
20	The description of maximum likelihood estimation could be improved. Saying “ML maximizes the likelihood of the observations given the unknown parameters” does not make it clear that the unknown parameters are variables in an optimization problem. Something like “ML is an optimization approach that determines the values of the unknown parameters that maximize the likelihood of the observed data occurring” makes the mechanism more apparent.	The description of the ML estimation has been improved in the first paragraph on page 16	16
21	What were the “groundbreaking contributions of Norbert Wiener”? Give a brief, simple description of the key contributions, so it is clear how they set up the development of the Kalman filter.	The description has been revised in Section 2.2.1, paragraph 1, to describe Wiener’s key contributions.	17
22	Simulations: 1) This is the bulk of the experimental work of the project. The tests are well-executed, but more explanation of some important aspects of the method is needed: How was the target simulated? This is explained to some extent later in the report with the different aircraft movement scenarios, but it might be better to describe these test cases here and then identify which one is used in the preliminary parameter comparisons, as I was left wondering where the ground truth came from while reading.	The scenario for the preliminary parameter comparisons has been expanded to clearly describe the simulation setup. Figure 4.4 has been added to illustrate the aircraft movement and corresponding ground truth used in the simulations.	75
23	Likewise, what is the ground truth for the MTT tests? It was not clear to me what the correct paths of the various targets	The ground truth for the MTT tests has been added to show what the various targets were expected to look like.	97

	were supposed to look like, and therefore, whether they were reproduced accurately.	This is illustrated in Figures 4.24 and 4.25, where the ground truth and aircraft movements are shown.	
24	The tests demonstrating the effects of different filter parameters were useful for establishing which parameters had an effect on the output, but it is difficult to quantify the sensitivity from looking at the plots. Summarizing the effects in terms of a cumulative error metric would help.	Cumulative error metrics for the different filter parameters have been added to quantify the sensitivity to different parameters, in Figures 4.17, 4.19, 4.21, and 4.23.	90,91,92,93,94 and 95
25	It would be helpful to have a similar experiment for the parameters of the CA-CFAR (especially the P_{fa} parameter), as it is not explained how these values were selected. Particularly, a value of $P_{fa} = 1 \times 10^{-5}$ is listed in table 4.2, but figures 4.10 and 4.11 reference a value of $P_{fa} = 1 \times 10^{-7}$. Were these selected by trial and error? If so, how sensitive were the results to these changes?	The CFAR validation figure in 4.12 has been updated to display the detection threshold for multiple P_{fa} values, illustrating the sensitivity of the threshold to this parameter. An accompanying paragraph has been added on page 81 paragraph 1 explaining the selection rationale, where a systematic sensitivity analysis is used to justify the chosen value.	81,82
26	The same as the above can be said for the bandwidth parameter $\sigma = [5 \text{ Hz}, 10 \text{ km}]$ of the clustering algorithm. How was this selected and how sensitive were the results to this value? Again, be careful of very small fonts on the figures (E.g. fig 4.13)	The bandwidth parameter was selected empirically through visual inspection of the clustering output. While a fixed bandwidth is an inherent limitation of the mean-shift algorithm, given the varying target extent in the ARD map, the chosen values ensure consistent tracking performance throughout the trajectory. The plot in Figure 4.16 (previously 4.13) has also been improved.	90
27	Performance Evaluation of Tracking Filters: 1) As in the previous section, the tests are well-executed and	A discussion defining the real-time processing threshold for the prediction and update stages has been added to the Processing Time subsection on pages 105–106. This provides context for interpreting Figures 5.4 and 5.5, clarifying when prediction times may become unsuitable for real-time application	105 and 106

	look at many aspects of the filters, but there is room for minor improvements in how these results are reported. • Figure 5.4, 5.5: It would be beneficial to have some additional context to interpret these results, namely, what is the threshold above which prediction time becomes unsuitable for real-time application? While it is very clear that the particle filter has a much longer prediction time, without the frame of reference of how long this time can be, it is unclear whether this difference is meaningful in a practical sense.		
28	• Data Association Performance: again – would be easier to interpret with more information about the reference targets for the MTT scenario.	The ground truth and target paths for the MTT scenario have been referenced to aid interpretation, as described in the paragraph below figure 5.21 on page 120.	120
29	• Tables 5.7 to 5.9: it is not easy to pull out a clear result by just reading the numbers, so an additional visual stimulus would help to make these instantaneously interpretable. I would suggest highlighting the best performing result or results in each row with one colour, and the worst result or results with another, so you can see immediately which column tends to include more good or bad results.	A visual stimulus has been added, along with a description in the section, to detail the different visual cues for Tables 5.8, 5.9, and 5.10.	121 and 122
30	It is critical to have an overarching summary of all results across the various metrics and scenarios. Something as simple as a colour-coded table, ranking the performance of all filters across the different tests performed e.g. would help to identify which filters performed best in particular contexts or metrics. This should then be used to make explicit recommendations about what the ultimate design should be, possibly with certain priorities in mind (e.g. what is the best approach if accuracy is critical? What is the best approach for minimizing load on hardware?)	A table providing an overarching summary across the various metrics and scenarios has been added as Table 5.3 in page 118. Scenarios have been included, and a discussion providing explicit recommendations has been added in the paragraph above the table and in the second paragraph in the conclusions on page 122.	118,122 and 123



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